# 2 Digital literacy: Searching the world wide web

## You will learn

- how to find information on the web and describe the sources you use
- how to choose information you find on the web, and give reasons for your choices
- how web search engines select and show useful information.

Last year you learned how to use a search engine to find information on the web. Some of the information you find on the web will be correct and reliable. Some of it will be incorrect and misleading. In this unit you will learn how to choose the best information for you.



### Talk about...

How do you learn about what is happening in the world? Do you use the web to find news? Can you trust what you read and watch on the web?





**Learning outcomes:** Obtain information from online sources and describe the sources used; Choose information from online sources, and give reasons for choices; Explain how online searches select and show useful information



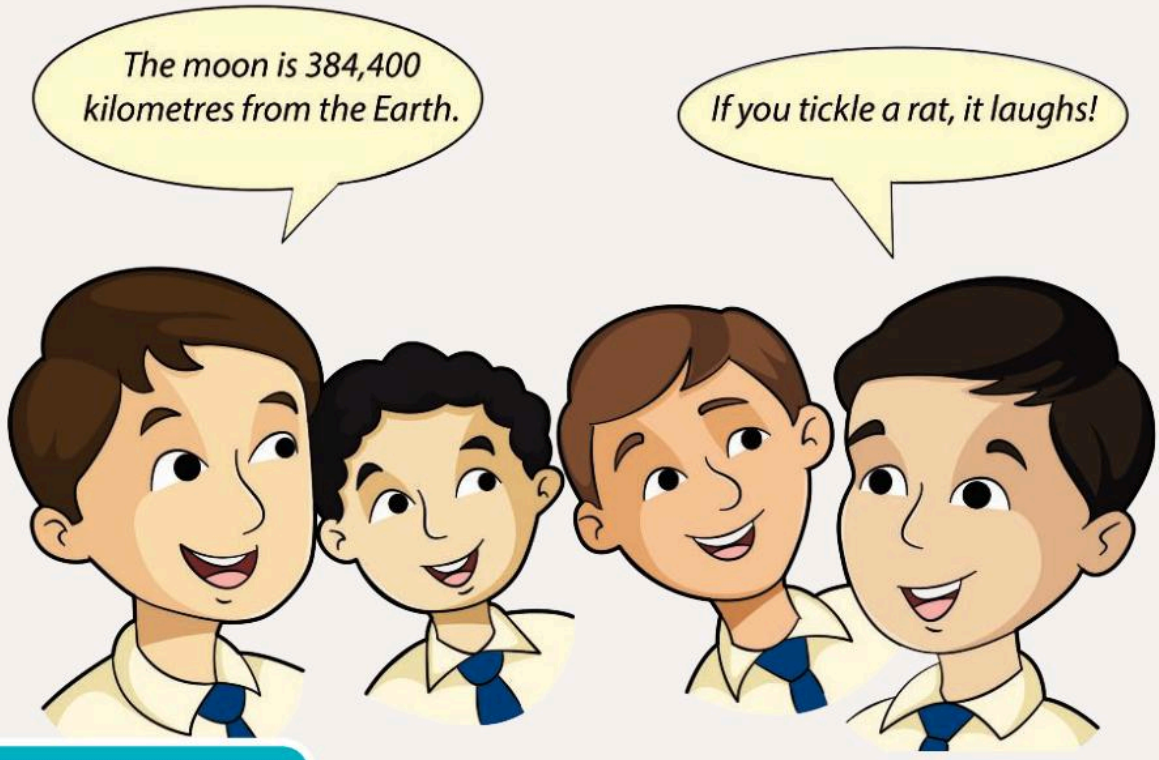
## Class activity

key word
search engine bookmark
spider web crawler meta tag
the web web page website
copyright

What is the most interesting fact that you have learned this week? Your fact can be funny. It can be sad. It can be about a big world event or about something that happened in your town. How did you find out this fact?

Discuss your facts in a small group. Make a list of the facts that your group thinks are the most interesting. Make sure you include one fact from each person in the group.

Do you believe all of the facts are true? Share your findings with the rest of the class.




## Did you know?

The world's first **search engine** was invented in 1990 by Alan Emtage. Alan Emtage was born in Barbados but moved to Montreal, Canada, to study at McGill University. His job at the university was to help other students find articles on the internet. He wrote a search engine to save himself time. He called his search engine Archie.

# 2.1 Searching the web

## In this lesson

You will learn:

- four golden rules to help you to search the web.

**Spiral back**

Last year you learned how to carry out basic world wide web searches. You learned how to use a search engine to find information quickly on the web. In this lesson you will review the search techniques that you learned last year.

## Four golden rules

### 1 Choose the right key words

Make sure you understand the question you need to answer. Before you enter anything into your search engine:

- Make a list of **key words**.
- Underline the most important key words.

Make sure that the question you type in the search engine contains the important key words. Your question does not need to include punctuation or short joining words like 'is' and 'the'.

### 2 Choose the right search engine

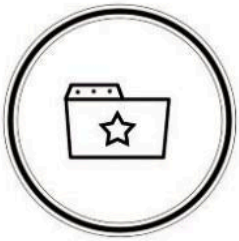
There are many search engines that you can use. Google, Bing and Yahoo are examples. Try using different search engines. Find out which one works best for you.

Some search engines are designed for young people. The advantages of using a child-friendly search engine are:

- The links are easier for you to read and understand.
- You are less likely to see unsuitable web pages.
- There are fewer adverts.



### 3 Bookmark your favourite sites



Adding a **web page** to your **bookmarks** makes it easy to find next time you want to use it. You don't have to search for the page again. You click on the link in your bookmarks list and go straight to the page.

Your bookmarks can be a good place to start a search. **Websites** you have found useful in the past might provide the information you need.

### 4 Try again

Sometimes a search does not give you the information you need. If that happens, try again.

- Try to think of different key words.
- Will a different search engine give better results?
- Will any of your bookmarked pages lead you to helpful information?



## Activity

Use a search engine to answer these questions.

1. What is Tungurahua, and where is it?
2. What happened at Tungurahua in March 2016?
3. Find a picture of Tungurahua.
4. Find one interesting fact about Tungurahua.
5. How do you pronounce Tungurahua?

**Think again**

Work in a small group. Discuss websites that you have found useful in your studies. Make a list of the best websites you have found. Why do you like them? What subjects do these websites help with? Present your list to the class.



## Extra challenge

Search the web to find the name of this bird.

**Hint:** To get a list of key words, type the things you see in the picture, such as 'black head'. What else do you see? Do an image search and find a picture that matches this one.

Which combination of key words worked best?



# 2.2 How search engines work

## In this lesson

You will learn:

- how a search engine finds web pages to answer your search questions.



## How large is the web?

At the start of 2019 there were 1.9 billion (1,900,000,000) websites on the web. It is easy to imagine what a small number like 10 or 100 looks like. It is more difficult to imagine what a large number like 1 billion looks like.

Think about this book:

- This lesson contains 500 words.
- The complete book contains 25,000 words.
- A pile of 76,000 books would contain 1.9 billion words. That is the same as the number of websites on the web.
- The pile of books would be 400 metres tall.
- That is half the height of the world's tallest building, Burj Khalifa.

## How does a search engine work?

When you enter a web search, your question is sent over the internet to a search engine. The search engine uses a powerful computer to look for matches to your key words. A list of links is displayed on your screen.

A search engine would take a very long time to search all 1.9 billion websites on the web. To be able to send you a list of links quickly, the search engine must work in a different way.

### Spiders and web crawlers

A search engine uses **web crawling** to search the web. The search engine sends a piece of software called a **spider** to crawl around the web and collect information.

When a spider reaches a web page, it records every word on the page and counts how many times each word is used. Then the spider follows links on the web page and carries out a word count on each new page it finds. Eventually, the spider will visit and record every web page on the web.



### Search engine index

The spider software sends the information it collects back to the search engine. The information is stored in a special list called an **index**. When you enter a search question, the search engine searches its index. The index is very large, but searching it is much quicker than searching the whole web.

A search engine compares each key word in your search question to the words stored in its index. It ignores short joining words like 'the', 'what' and 'why'. These words appear on every web page so won't help in a search.

The search engine uses an **algorithm** to compare websites in its index with the key words in your web search. The algorithm chooses the web pages to answer your question.



## Activity

Copy this search question: 'What is the longest river in Africa?'

Draw circles around all the words that the search engine will ignore. What is the answer to the question?



## Extra challenge

A search engine uses an algorithm to choose websites to answer a web search question. What is an algorithm? Use the knowledge you have already gained in this course. Search the web for more information if you need to.

## Think again

Carry out a web search to find a video or animation that explains how a search engine works. Watch the video and make notes. Bookmark the video that you find most useful.